

Assignment 2; due Thursday, October 8, 2026

- [1] Let  $(\mathbf{x}_k)_{k \in \mathbb{N}}$  be a sequence in  $\mathbb{R}^n$ . Write  $\mathbf{x}_k = (x_{k1}, \dots, x_{kn})$ , so that  $(x_{kj})_{k \in \mathbb{N}}$  is a sequence in  $\mathbb{R}$  for each  $j = 1, \dots, n$ .
- [a] Let  $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n$ . Prove that  $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{a}$  if and only if  $\lim_{k \rightarrow \infty} x_{kj} = a_j$  for each  $j = 1, \dots, n$ . (*Don't forget to prove both directions!*)
- [b] Prove that  $(\mathbf{x}_k)_{k \in \mathbb{N}}$  is a Cauchy sequence in  $\mathbb{R}^n$  if and only if  $(x_{kj})_{k \in \mathbb{N}}$  is a Cauchy sequence in  $\mathbb{R}$  for each  $j = 1, \dots, n$ . (*Don't forget to prove both directions!*)

- [2] The two parts of this problem *are* related.
- [a] Let  $E \subseteq \mathbb{R}$  be nonempty. If  $E$  is bounded below, so that  $\inf E$  exists, prove that there exists a sequence  $(a_k)$  with  $a_k \in E$  for all  $k$ , and  $\lim_{k \rightarrow \infty} a_k = \inf E$ .
- [b] Let  $A \subseteq \mathbb{R}^n$  be nonempty, and  $\mathbf{x} \in \mathbb{R}^n$ . We define the *distance* from  $A$  to  $\mathbf{x}$  to be

$$\text{dist}_A(\mathbf{x}) = \inf\{\|\mathbf{x} - \mathbf{y}\| : \mathbf{y} \in A\}.$$

(Note that  $\text{dist}_A(\mathbf{x})$  is well-defined and  $\text{dist}_A(\mathbf{x}) \geq 0$ , because it is the infimum of a nonempty set of nonnegative real numbers.) Prove that there exists a sequence  $(\mathbf{y}_k)$  in  $\mathbb{R}^n$  with  $\mathbf{y}_k \in A$  for all  $k$  and  $\lim_{k \rightarrow \infty} \|\mathbf{x} - \mathbf{y}_k\| = \text{dist}_A(\mathbf{x})$ .

- [3] Let  $A \subseteq \mathbb{R}^n$ . We define the *closure* of  $A$ , denoted by  $\overline{A}$ , to be the subset

$$\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap A \neq \emptyset \text{ for all } \varepsilon > 0\}.$$

That is, a point  $\mathbf{x} \in \mathbb{R}^n$  lies in the closure of  $A$  if and only if every open ball of any positive radius intersects  $A$ .

- [a] Show that  $A \subseteq \overline{A}$ .
- [b] Show that  $\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : U \cap A \neq \emptyset \text{ for any open set } U \text{ containing } \mathbf{x}\}$ .
- [c] Show that  $\overline{A}$  is closed.  
*Hint:* First show there exists  $\varepsilon_0$  such that  $B_{\varepsilon_0}(\mathbf{x}) \cap A = \emptyset$ , and then show that  $B_{\varepsilon_0}(\mathbf{x}) \cap \overline{A} = \emptyset$ . You will need to use part [b].
- [d] If  $A$  is closed, show that  $\overline{A} = A$ .
- (Combining parts [c] and [d] shows that  $A$  is closed if and only if  $\overline{A} = A$ .)

- [4] Let  $E \subseteq \mathbb{R}^n$  be nonempty.
- [a] If  $(\mathbf{x}_k)_{k \in \mathbb{N}}$  is a sequence such that  $\mathbf{x}_k \in E$  for all  $k \in \mathbb{N}$ , and  $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{p}$ , prove that  $\mathbf{p} \in \overline{E}$ .
- [b] Let  $\mathbf{p} \in \overline{E}$ . Prove that there exists a sequence  $(\mathbf{x}_k)_{k \in \mathbb{N}}$  such that  $\mathbf{x}_k \in E$  for all  $k \in \mathbb{N}$ , and  $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{p}$ . *Hint:* Consider  $B_{\frac{1}{k}}(\mathbf{p})$ .

Taken together, these two statements say that  $\overline{E}$  is exactly the set of limits of convergent sequences whose elements all lie in  $E$ .

- [5] Let  $(\mathbf{x}_m)_{m \in \mathbb{N}}$  be a sequence in  $\mathbb{R}^n$ . Define a new sequence  $(\mathbf{s}_k)_{k \in \mathbb{N}}$  by

$$\mathbf{s}_k = \sum_{m=1}^k \mathbf{x}_m.$$

If the sequence  $(\mathbf{s}_k)$  converges, then we say that *the series*  $\sum_{m=1}^{\infty} \mathbf{x}_m$  *converges*. If  $\lim_{k \rightarrow \infty} \mathbf{s}_k = \mathbf{s}$ , then we write  $\sum_{m=1}^{\infty} \mathbf{x}_m = \mathbf{s}$ .

- [a] Prove that  $\sum_{m=1}^{\infty} \mathbf{x}_m$  converges iff: for all  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$\left\| \sum_{m=k+1}^{\ell} \mathbf{x}_m \right\| < \varepsilon$$

for all  $\ell \geq k \geq N$ .

*Hint:* A sequence in  $\mathbb{R}^n$  converges if and only if it is Cauchy.

- [b] Suppose that  $\sum_{m=1}^{\infty} \mathbf{x}_m$  converges. For any  $N \in \mathbb{N}$ , prove that  $\sum_{m=N}^{\infty} \mathbf{x}_m$  converges, and that

$$\lim_{N \rightarrow \infty} \left( \sum_{m=N}^{\infty} \mathbf{x}_m \right) = 0.$$

*Hint:* Consider the sequence  $(\tilde{\mathbf{s}}_{\ell})_{\ell \geq N}$  given by  $\tilde{\mathbf{s}}_{\ell} = \sum_{m=N}^{\ell} \mathbf{x}_m$ .

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- [6] Let  $(\mathbf{x}_k)_{k \in \mathbb{N}}$  be a sequence in  $\mathbb{R}^n$ . Given  $\mathbf{x} \in \mathbb{R}^n$ , prove that  $(\mathbf{x}_k)$  has a *subsequence* converging to  $\mathbf{x}$  if and only if, for all  $\varepsilon > 0$  the open ball  $B_{\varepsilon}(\mathbf{x})$  contains  $\mathbf{x}_k$  for infinitely many  $k$ .

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- [7] We know that a nonconvergent sequence can have subsequences which can converge to *different* limits. Give an example of a sequence in  $\mathbb{R}^n$  which has *no convergent subsequences* at all. Justify your answer. *Hint:* Recall the Bolzano–Weierstrass Theorem.

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- [8] Recall that a subset  $K$  of  $\mathbb{R}^n$  is called *compact* iff every sequence in  $K$  admits a subsequence that converges to a point in  $K$ .

Let  $K_1, K_2, \dots, K_m$  be a finite collection of compact subsets of  $\mathbb{R}^n$ . Give two different proofs that the union  $\cup_{j=1}^m K_j$  is compact:

[a] using the definition of compactness.

[b] using the Heine–Borel theorem.

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- [9] Let  $E \subseteq \mathbb{R}^n$  be nonempty. For  $\mathbf{x} \in \mathbb{R}^n$ , we define the *distance* from  $E$  to  $\mathbf{x}$ , denoted  $\text{dist}_E(\mathbf{x})$ , to be

$$\text{dist}_E(\mathbf{x}) = \inf \{ \|\mathbf{x} - \mathbf{z}\| : \mathbf{z} \in E \}.$$

Thus we can think of  $\text{dist}_E$  as a function from  $\mathbb{R}^n$  to  $\mathbb{R}$ .

[a] Prove that  $|\text{dist}_E(\mathbf{x}) - \text{dist}_E(\mathbf{y})| \leq \|\mathbf{x} - \mathbf{y}\|$  for all  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ .

[b] Prove that  $\text{dist}_E: \mathbb{R}^n \rightarrow \mathbb{R}$  is continuous on  $\mathbb{R}^n$ .

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- [10] Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be given by

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q}, \\ 1 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Prove that  $f$  is not continuous at  $x$  for any  $x \in \mathbb{R}$ .

*Hint:* Find a way to use the sequential characterization of continuity.

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